

Heaven's light is our guide



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Mechatronics Engineering

Lab Report

Experiment no.:07

Experiment name: Study On the Various Real-World Problem Solved Using Basic Logic in Ladder Diagram Programming.

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Date of Experiment: 23 August, 2025

Date of Submission : 20 September ,2025

MTE 4102

Automation Sessional

Experiment no.:07

Experiment name: Study On the Various Real-World Problem Solved Using Basic Logic in Ladder Diagram Programming.

Objectives:

- To understand the use of timers, counters, and interlocks in solving real-world automation problems using ladder diagram programming.
- To design and simulate logic control systems such as traffic light control, batch mixing, washing machine cycle, and elevator operation.
- To develop problem-solving skills by applying sequential logic for safe, efficient, and automated industrial and domestic applications.

Theory:

Ladder Diagram (LD) is one of the most widely used PLC programming languages in industrial automation. It represents electrical relay logic in symbolic form, where rungs of the ladder define the logical sequence of operation. For solving real-world problems, basic instructions like contacts, coils, timers (TON/TOF), and interlocks are applied.

Timers help in creating delays or maintaining a task for a fixed period.

Interlocks ensure safety by preventing conflicting operations such as preventing motor start before door closure in elevators or preventing car and pedestrian lights from turning green simultaneously in traffic control.

Sequential logic ensures step-by-step execution of tasks like filling, mixing, or spinning in washing machines and batch mixing systems.

This structured approach allows for error-free, repeatable, and safe operation of machines, making LD programming essential in industries and automation systems.

Required Apparatus:

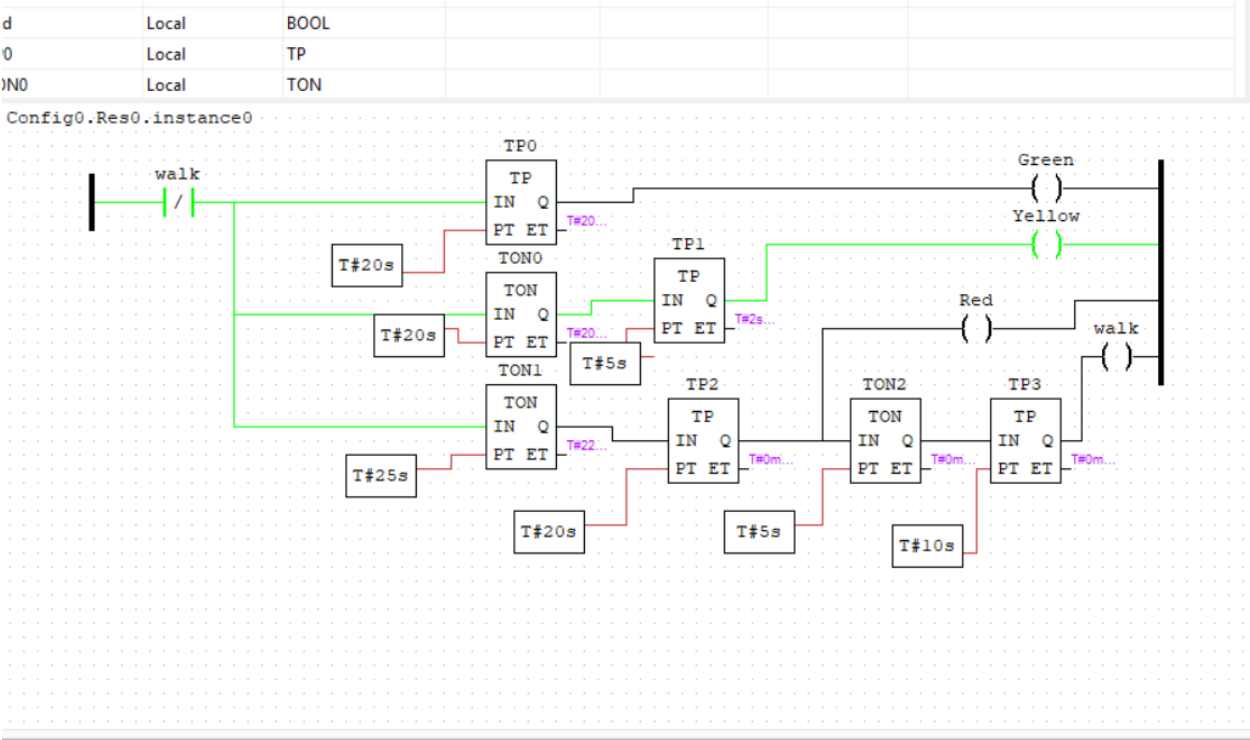
1. Laptop
2. OpenPLC Editor (PC Software)

Problem Statements & Ladder Diagram:

Problem1: Traffic Intersection with Pedestrian Crossing

Two perpendicular roads (North-South & East-West), and each direction has Green → Yellow → Red sequence with timing Green: 20s, Yellow: 5s, Red: 20s. If pressed during Red, pedestrian signal (Walk) turns ON for 10s. Pedestrian signal must not interrupt car cycle. Please use timers and interlocks to ensure no conflicts.

Ladder Diagram:



Problem2: Batch Mixing System

In a chemical processing industry, there are three tanks (A, B, C) must be discharged into a mixer. The sequence of batch mixing are firstly, open valve A for 5s. After 2s pause, open valve B for 8s. After 2s pause, open valve C for 10s. Then after all valves close, mixer motor runs for 20s. Finally, when mixing is complete, a buzzer should turn ON for 3s.

[illegible]

The automatic washing machine has gone to complete the cycle sequence using timers: Fill water (10 sec), then Agitate (15 sec), then Drain (5 sec) and finally Spin (10 sec). After all tasks are complete, a buzzer should turn ON for 3s.

g: Config0.Res0.instance0

start

fill water

Agitate

Drain

spin

Buzzer

Problem4: Elevator Simulation

Elevator motor controlled by Up/Down pushbuttons. If UP is pressed, firstly the door closes (3s) and then Motor runs Up for 10s for each floor. Finally, door opens (3s). If no person is detected for 20 seconds, then automatically stop. If DOWN is pressed, the Same sequence. Please provide the safety interlock so that the door must always close before motor starts.

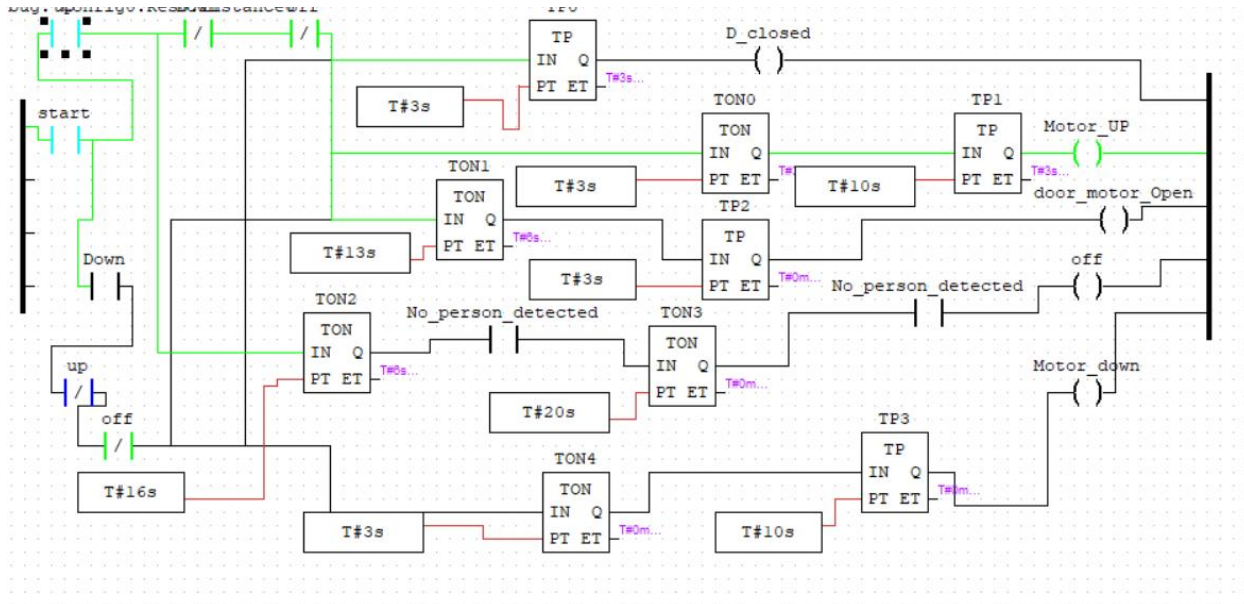


Fig: when up button pressed & down off

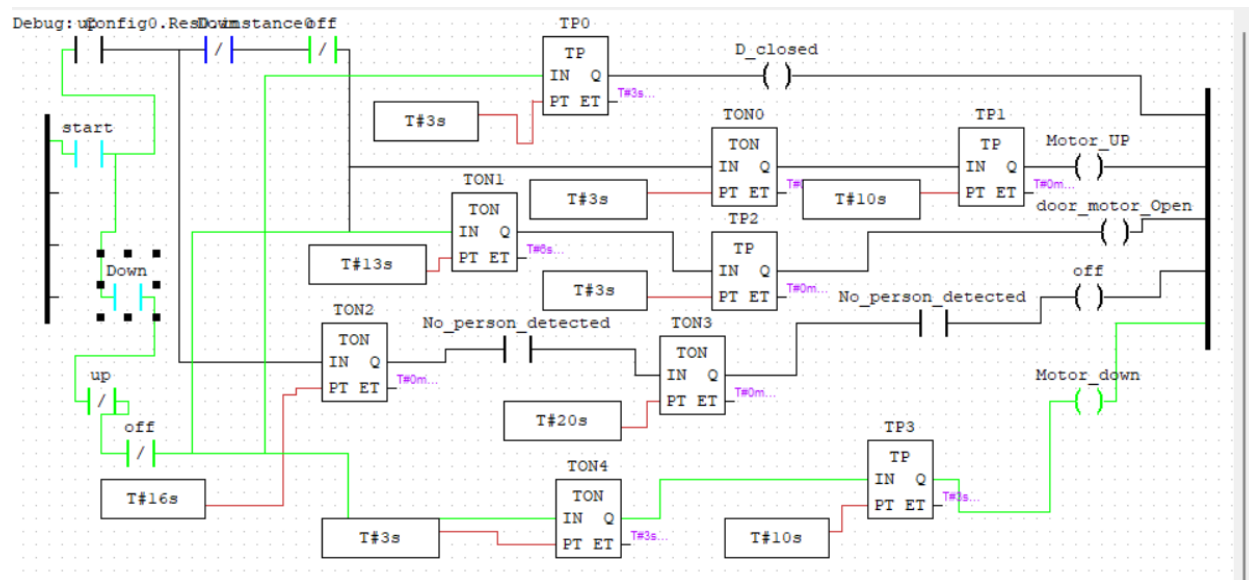


Fig: when down button pressed & up off

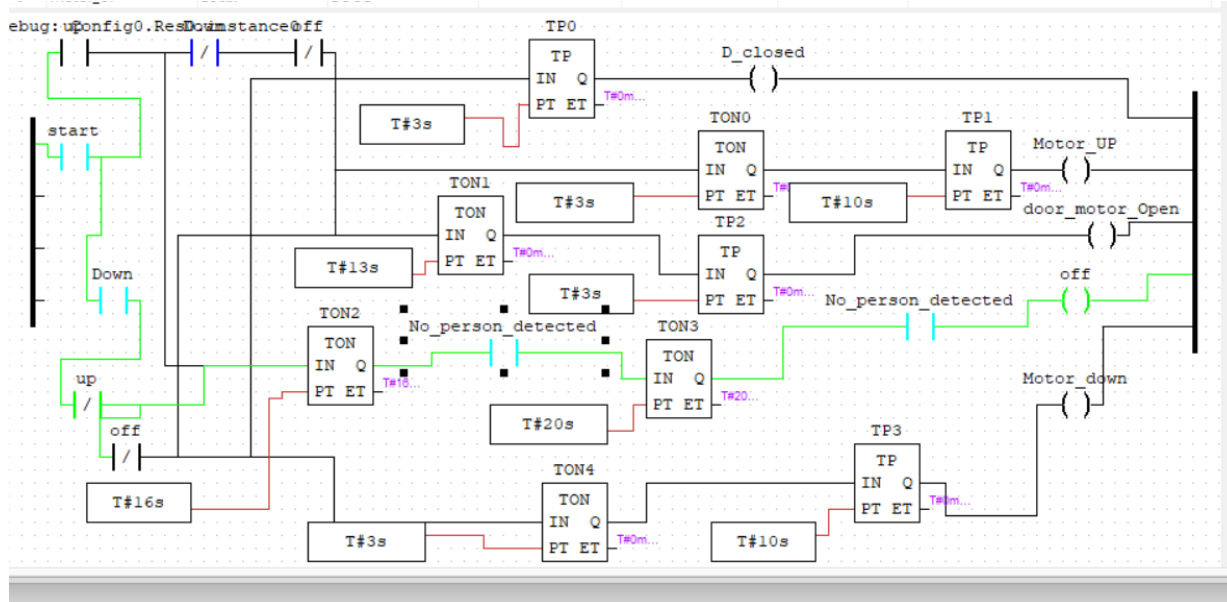


Fig: When no person detected for 20 sec

Discussion:

In this experiment, different real-life automation problems were modeled using ladder logic. The traffic intersection with pedestrian crossing demonstrated how timers and interlocks can manage light cycles while ensuring pedestrian safety without interrupting the vehicle sequence. The batch mixing system highlighted industrial process control, where proper sequencing and pauses between valve operations ensured correct material proportions before mixing. The automatic washing machine cycle reflected how domestic appliances use timers to complete multi-step tasks efficiently. Finally, the elevator simulation showed the importance of safety interlocks, ensuring that the door always closes before motor movement, and automatic stopping after inactivity for energy efficiency and safety. These cases collectively show how ladder diagrams can translate practical needs into logical automation solutions.

Conclusion:

The lab successfully demonstrated how basic logic design using Ladder Diagrams can solve real-world problems in traffic management, industrial batch processing, household appliances, and elevators. By using timers, interlocks, and sequential operations, reliable and safe automation sequences were created. This emphasizes the importance of ladder programming in both industrial and domestic applications, making it a fundamental skill for engineers working with automation and control systems.

References:

- [1]. Frank D. Petruzella, Programmable Logic Controllers, 4th Edition, McGraw-Hill, 2017.
- [2]. John W. Webb and Ronald A. Reis, Programmable Logic Controllers: Principles and Applications, Prentice Hall, 2015.
- [3]. OpenPLC Project, "OpenPLC Official Website." [Online]. Available: <https://www.openplcproject.com/>
- [4]. IEC 61131-3 Standard, Programmable Controllers – Programming Languages, International Electrotechnical Commission.